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Alireza Ghorbani

Curriculum Vitae

September 26, 2023

Education

2014–2016 **M.Sc. in Mathematical Statistics**, Shahid Beheshti University
- *Thesis: Analysis of High Dimensional Data Using Rayleigh Quotient Optimization (in Persian) under supervision of Prof. Faridrohani & prof. Hamedani, Grade: A (19/20)*
- *GPA: 16.10/20 (3.38/4)*

2009–2014 **B.Sc. in Statistics**, Shahid Chamran University
- *Thesis: Statistical Study for The Factors Contributing to The Academic Failure among A Group of Undergraduate Students (in Persian) under supervision of Prof. Zadkarami*
- *GPA: 13.95/20 (2.57/4) – Last Semester GPA: 16.83/20 (4/4)*

2006–2009 **High School Diploma in Mathematics and Physics**, *Shahid Beheshti High School*, Birjand, Iran
Affiliated with the National Organization for Development of Talented Students (NODET)
- *CGPA: 17.45/20 (3.83/4)*

Professional Experience

2021 - Now **Market Research Analyst & Statistician**, *R&D Department, Zarrin Roya*, Tehran, Iran

2022 – 2023 **Statistics & Machine learning Instructor**, *Learnix.ir*, Tehran, Iran

Publication

2023 **Optimizing Water Pipeline Route Selection Using a Sparse Deep Neural Network and the Fuzzy VIKOR Method (case study: water pipeline transmission)**, *Ghorbani, A. Khajehzadeh, M. Seifi, F. Al-Ansari, N and Shahabi, H.* , Preprint

Honors and Awards

2014–2016 **Awarded Fellowship**, *Graduate Program*, Statistics Department, Shahid Beheshti University, Tehran, Iran

2014 **Ranked 9th**, *in 19th National Scientific Olympiad (Statistics)*

2014 **Ranked 19th**, *in National Entrance Exam for Masters of Science in Actuarial Statistics, Social & Economic Statistics and Theoretical Statistics*

2009–2014 **Awarded Fellowship**, *Undergraduate Program*, Statistics Department, Shahid Chamran University, Ahwaz, Iran

2003 & 2005 Selected for study in schools of National Organization for Development of Exceptional Talents (NODET) through an exam with less than 1% acceptance rate.

Technical skills

R Programming, Python , Matlab, SQL, LaTeX

Research Interest

- Natural Language Processing
- High-Dimensional Statistics
- Statistical Learning
- Bayesian Statistics

Research Experience & Selected Projects

Comparison of Sentiment Analysis Methods for Farsi Text, Developing models (Logistic lasso Regression, Naive Bayes, Vector Space Model and LSTM) for sentiment analysis and text classification of Farsi text in , and comparing their performances. The best method selected for tracking customers' behavior over social media.

Recommender System Suggesting Proper Product Size, Developing a recommending system to help mothers find proper diaper size in . Selecting proper diaper size could be a challenge for young mothers that could lead to a significant amount of dissatisfaction.

Water Pipeline Optimization using Multi-Task Sparse Neural Network Integrated with MCDM, The costly intervention of five groups of experts that their idea regarding water pipeline was necessary for evaluation and determination of the best water pipeline was replaced by a multi-task neural network model. Finally, some newly proposed pipeline alternatives were qualified by the model and ranked by FVIKOR (developed in).

Sentiment Analysis Using Multiple Methods Based on Bag of Words Approach for Persian Text, *In this project, I classified Persian text by the polarity (positive or negative opinions) expressed within them using 3 different method in .*

High Dimensional Classification using Rayleigh Quotient, *I have studied classification methodologies and the challenges for their implementation facing high-dimensional data. Then, I have provided numerical results (by simulation and by an actual breast cancer dataset) comparing these methods. This project was a part of my master's degree thesis and was done in and .*

QUADRO, *The methodology that was introduced in the paper "QUADRO: A Supervised Dimension Reduction Method Via Rayleigh Quotient Optimization" developed and implemented in and .*

ROAD, *The code for the paper "A ROAD to Classification in High Dimensional Space" translated from to .*

Academic Failure Detection Model, I have investigated the relevant factors that can lead to academic failure in order to detect vulnerable students providing them the help and supervision. This project was a part of my bachelor's degree thesis.

Extracurricular Activities & Professional Certificates

- TensorFlow Developer Professional Certificate
  deeplearning.ai
- Natural Language Processing Specialization
  deeplearning.ai
- Bayesian Statistics Specialization
  University of California, Santa Cruz
- Statistics with Python Specialization
  University of Michigan
- PostgreSQL for Everyone Specialization
  University of Michigan
- Deep Learning Specialization
  deeplearning.ai
- Machine Learning Specialization
  Stanford University & deeplearning.ai
- Fundamentals of Reinforcement Learning
  University of Alberta
- Python 3 programming Specialization
  University of Michigan

Test Scores

July 2020 **Scored 103 in TOEFL iBT Exam**

Reading (25/30), Listening (29/30), Speaking (25/30), Writing (24/30)

Languages

- English: Fluent
- Spanish: Elementary
- Persian: Native